PHYSICAL CUME
SEPTEMBER 26, 2003

Phosphorus pentachloride is ionic in the solid phase but is a covalent molecule, PCl_5 in the gas phase. 0.1 mol (20.824 g) of phosphorus pentachloride is dropped into a 3 L vessel which is kept at 250° C and contains chlorine gas at 0.5 atm partial pressure. PCl_5 vaporizes and then partially dissociates under these conditions, according to:

$$PCl_5 \rightleftharpoons PCl_3 + Cl_2$$

The equilibrium constant $K_p = 1.78$ atm at 250° C.

(a) What is the degree of dissociation of PCl_5 (the percentage of PCl_5 dissociated)?

(b) What would be the total pressure if 0.05 mol (10.41 g) of PCl_5 is placed in a cold, evacuated 3 L vessel and then heated to 250° C?

The gas constant R=0.08206 L atm/(mol K). The solutions of the quadratic equation $ax^2+bx+c=0$ are given by $x=(-b\pm(b^2-4ac)^{1/2})/(2a)$